

Bo Xu

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PROFESSIONAL SUMMARY

Senior Data and AI Engineer with 15 years of experience architecting enterprise data infrastructure and building production-grade RAG and agentic AI systems. Proven track record of deploying RAG assistants over structured financial data and governance metadata at a major hedge fund, incorporating automated retrieval accuracy, faithfulness evaluation, and runtime guardrails. Currently Founding Engineer and CTO at Mach-AI, leading end-to-end architecture and agent orchestration workflows. Combines deep Python and SQL engineering with strong business and financial data fluency derived from actuarial analytics, hedge fund operations, and a Columbia University Master's in Computer Science.

CORE COMPETENCIES

RAG & Agentic AI Systems

Production SQL & Python Engineering

Data Modeling & Metadata Governance (Unity Catalog)

PySpark & Distributed Cloud Infrastructure

Self-Healing Automation & Pipeline Reliability

Data Contracts & Lineage Controls

Financial & Business Data Fluency

CI/CD (GitHub Actions) & Test Discipline

WORK EXPERIENCE

Mach-AI

February 2026 - Present

Founding Engineer & CTO

Los Angeles, CA

- Ship production agentic AI workflows using LLM APIs and agent orchestration, with evaluation discipline (accuracy checks, guardrails) - building the tools that let AI act on business data.
- Founding engineer owning the full stack - architecture, data infrastructure, integrations, deployment, and support - at an early-stage startup; work AI-first daily, continuously adopting new models and capabilities.
- Establish engineering standards (version control, CI/CD, testing with black, mypy, unit tests) and drive pragmatic build-vs-buy decisions.

Centiva Capital

September 2022 - February 2026

Senior Data Engineer

New York, NY

- Designed and shipped a production RAG system over structured business data - a knowledge-access assistant over data contracts and governance metadata - owning retrieval, embeddings, chunking, prompt strategy, and accuracy/faithfulness evaluation with fallback guardrails.
- Built LLM-driven parsing that turned unstructured documents and communications into structured, queryable data for business teams - with evaluation keeping outputs trustworthy in a regulated environment.
- Defined ground-truth data foundations: data contracts, governance and metadata (Unity Catalog), lineage, and access controls that made the firm's data reliable enough for AI and research workloads to build on.
- Built and operated 100+ production pipelines (Python, SQL, PySpark, Databricks on AWS) feeding research and ML workflows - with CI/CD, monitoring, and self-healing automation that treated manual intervention as a bug.
- Partnered directly with portfolio managers, analysts, and legal to turn ambiguous, revenue-critical questions into shipped data and AI products.

RIMES Technologies

March 2015 - September 2022

Senior Data Developer

New York, NY

- Customer-embedded delivery for 300+ global financial institutions: worked directly with client, product, and engineering teams to onboard and integrate 800+ heterogeneous data sources and keep mission-critical services flowing.
- Owned complex production escalations end-to-end - diagnosing issues across client environments and sustaining 95% timeliness against a 1-hour SLA.

- Mentored developers on production-readiness, reliability, and monitoring practices.

Anthem Blue Cross Blue Shield

January 2010 - May 2011

Actuarial Analyst

Portland, ME

- Built statistical and predictive models on health insurance data (SQL, Excel), producing analytics for senior leadership and supporting a record \$90B revenue quarter.

EDUCATION

M.S., Computer Science (Machine Learning) [Columbia University](#) 2014

B.S., Applied Mathematics & Economics [Columbia University](#) 2009